Flap Selector Valve – Maintenance Practices

1. Removal/Installation

   A. Remove Flap Selector Valve

      Refer to Figure 1 and Figure 2 for details.

      1) Remove forward cabin ceiling panels.

      2) Disengage HYDRAULIC OIL PUMP circuit breaker and install locking clip. Discharge hydraulic pressure from accumulators by operating brake pedals.

      3) Move flap selector lever to flap down position and pump flaps down, using handpump.

      4) Remove flap actuator forward drip tray.

      5) Slacken flap selector cable at turnbuckles. Identify up and down cables and remove cable terminals from selector valve selector pulley.

      6) Slacken selector valve follow-up cable at A-frame pulley adjusting nut, and remove cable terminals from selector valve follow-up pulley.

      7) Position suitable container and disconnect hydraulic lines at selector valve. Cap lines and ports.

      8) Support selector valve and remove four bolts and washers securing selector valve to flap actuator. Remove selector valve from aircraft.

   B. Install Flap Selector Valve

      Refer to Figure 1 and Figure 2 for details.

      1) Position selector valve on flap actuator and secure with four bolts and washers. Lockwire bolts.

      2) Position suitable container and remove caps from hydraulic lines and selector valve ports. Connect hydraulic lines to selector valve.

      3) Install follow-up cable terminals in selector valve follow-up pulley. Lockwire terminals in position.

         Note If selector valve pulleys are in flaps up position (refer to Figure 2), rotate to permit cable terminal installation.

      4) Install flap selector cable terminals in selector valve selector pulley. Lockwire terminals in position. Refer to Note in Para 1.B.3)).

         Note Flaps down selector cable is routed over selector pulley.

      5) Tension flap selector and follow-up cables. (Refer to 27-00-00, Flight Controls – General Data).

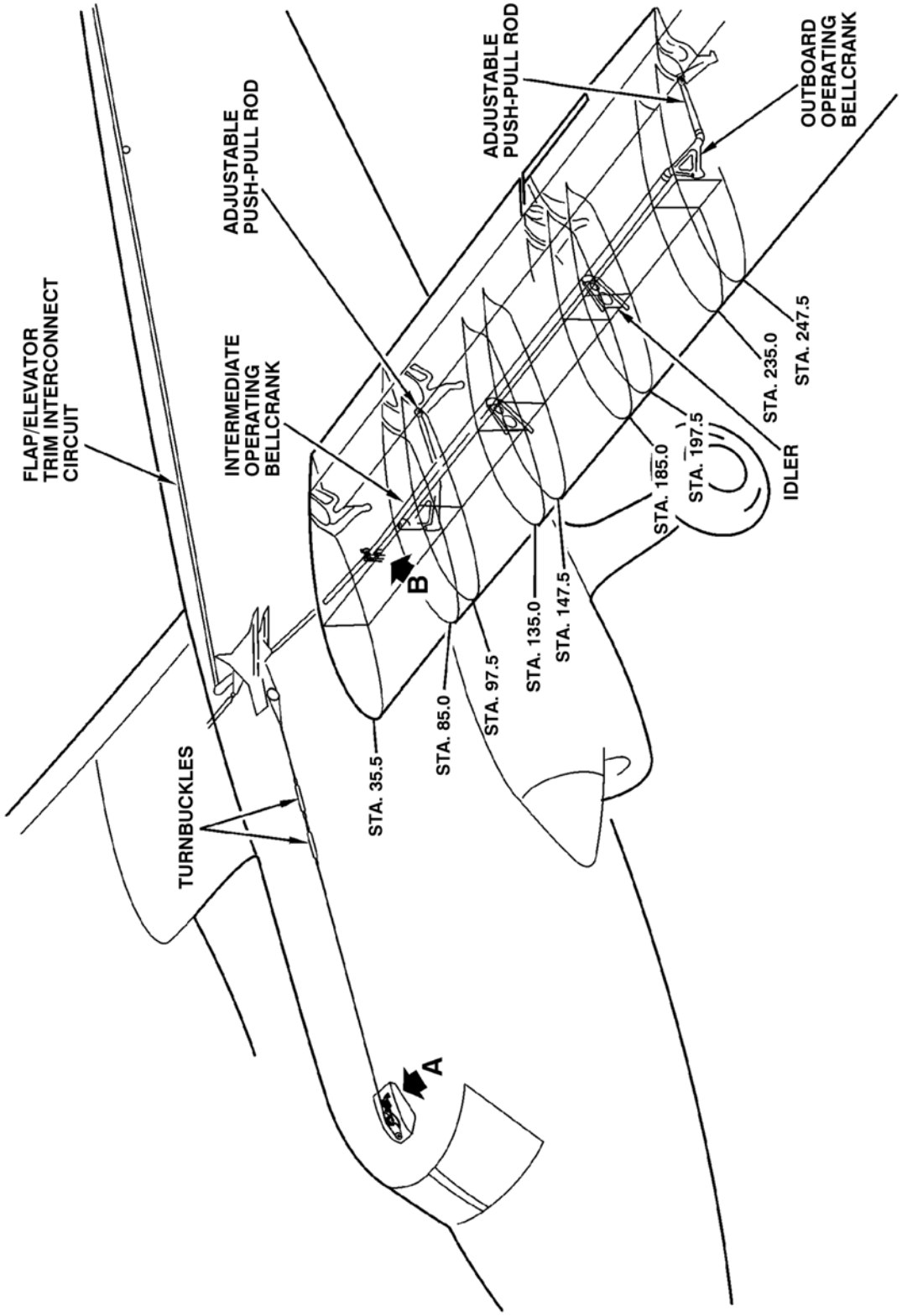
      6) Rig flaps. (Refer to 27-50-00, Wing Flaps System – General Data).

      7) Remove clip from HYDRAULIC OIL PUMP circuit breaker and bleed flap actuator. (Refer to 27-50-11, Flap Actuator – General Data).

      8) Do an operational test on wing flaps. (Refer to 27-50-00, Wing Flaps System – General Data).

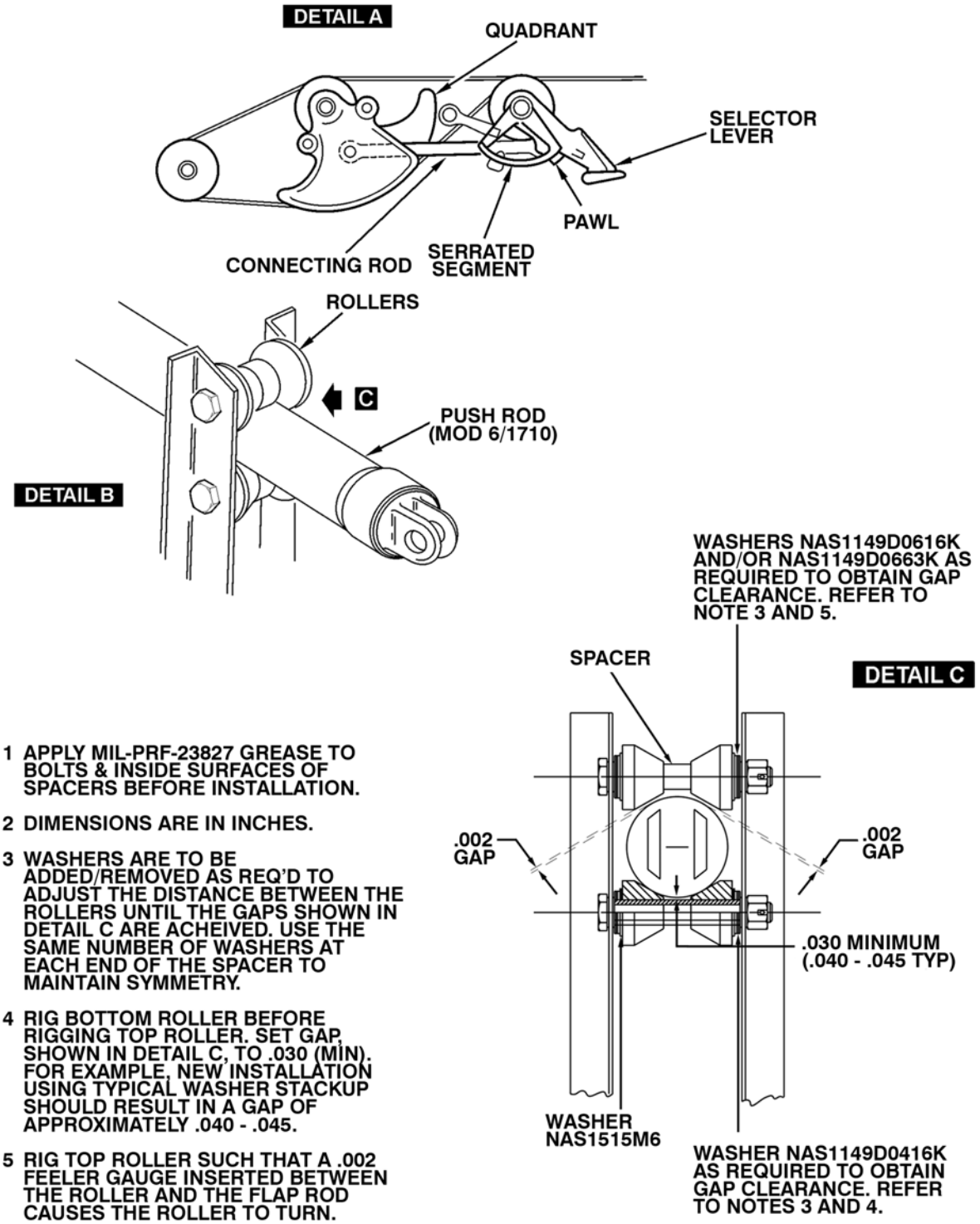
      9) Install flap actuator forward drip tray.

      10) Install forward cabin ceiling panels.



Wing Flap System
Figure 1 (Sheet 1 of 4)

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NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WASHERS ARE TO BE ADDED/REMOVED AS REQ'D TO ADJUST THE DISTANCE BETWEEN THE ROLLERS UNTIL THE GAPS SHOWN IN DETAIL C ARE ACHEIVED. USE THE SAME NUMBER OF WASHERS AT EACH END OF THE SPACER TO MAINTAIN SYMMETRY.

4 RIG BOTTOM ROLLER BEFORE RIGGING TOP ROLLER. SET GAP, SHOWN IN DETAIL C, TO .030 (MIN). FOR EXAMPLE, NEW INSTALLATION USING TYPICAL WASHER STACKUP SHOULD RESULT IN A GAP OF APPROXIMATELY .040 - .045.

5 RIG TOP ROLLER SUCH THAT A .002 FEELER GAUGE INSERTED BETWEEN THE ROLLER AND THE FLAP ROD CAUSES THE ROLLER TO TURN.

6 BOLTS SHOULD NOT BE OVER TIGHTENED. ROLLERS/SPACERS SHOULD TURN FREELY ON BOLTS.

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Wing Flap System
Figure 1 (Sheet 2 of 4)

NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WITH ROD PUSHED ALL THE WAY INSIDE THE WING (I.E. FLAP FULLY RETRACTED). PRE-FIT ROLLERS, SPACERS, & BOLTS (NOTE: NO WASHERS) BY CENTERING THE ROD ON ROOT RIB CUT-OUT AND BETWEEN ROLLERS/SPACERS. SPACERS SHALL BE APPROX CENTERED BETWEEN SUPPORT ANGLES.

4 SLIDE ROLLERS TO MAKE CONTACT WITH ROD, THEN MEASURE GAPS G1, G2, G3 & G4. REMOVE PRE-FIT ROLLERS, SPACERS & BOLTS.

5 DETERMINE QUANTITY OF NAS1149D0663K AND NAS1149D0616K WASHERS USING DIMENSION: Gx-.05 = Tx; CHOOSE EQUAL OR NEXT GREATER THICKNESS.

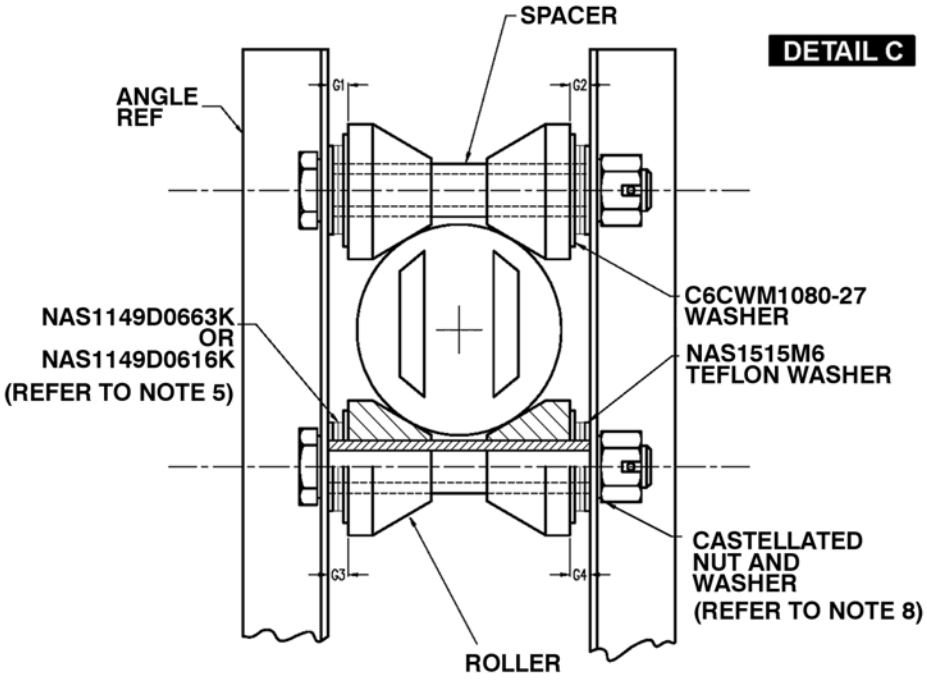
EXAMPLE:
G1=.189, THEREFORE G1-.05 = .139
REQD WASHER: 2x NAS1149D0663K = .126
1x NAS1149D0616K = .016
.142

6 INSTALL BOTTOM ROLLER/SPACER WITH WASHERS AS PER NOTE 5. ENSURE THE SPACER IS APPROX CENTERED: I.E. EQUAL GAP BETWEEN FWD & AFT SUPPORT ANGLE.

7 WHILE PUSHING THE FLAP ROD DOWNWARD TO COMPRESS THE LOWER ROLLERS, INSTALL THE UPPER ROLLERS/SPACER AS PER NOTE 5. CENTER THE SPACER ON THE FLAP ROD.

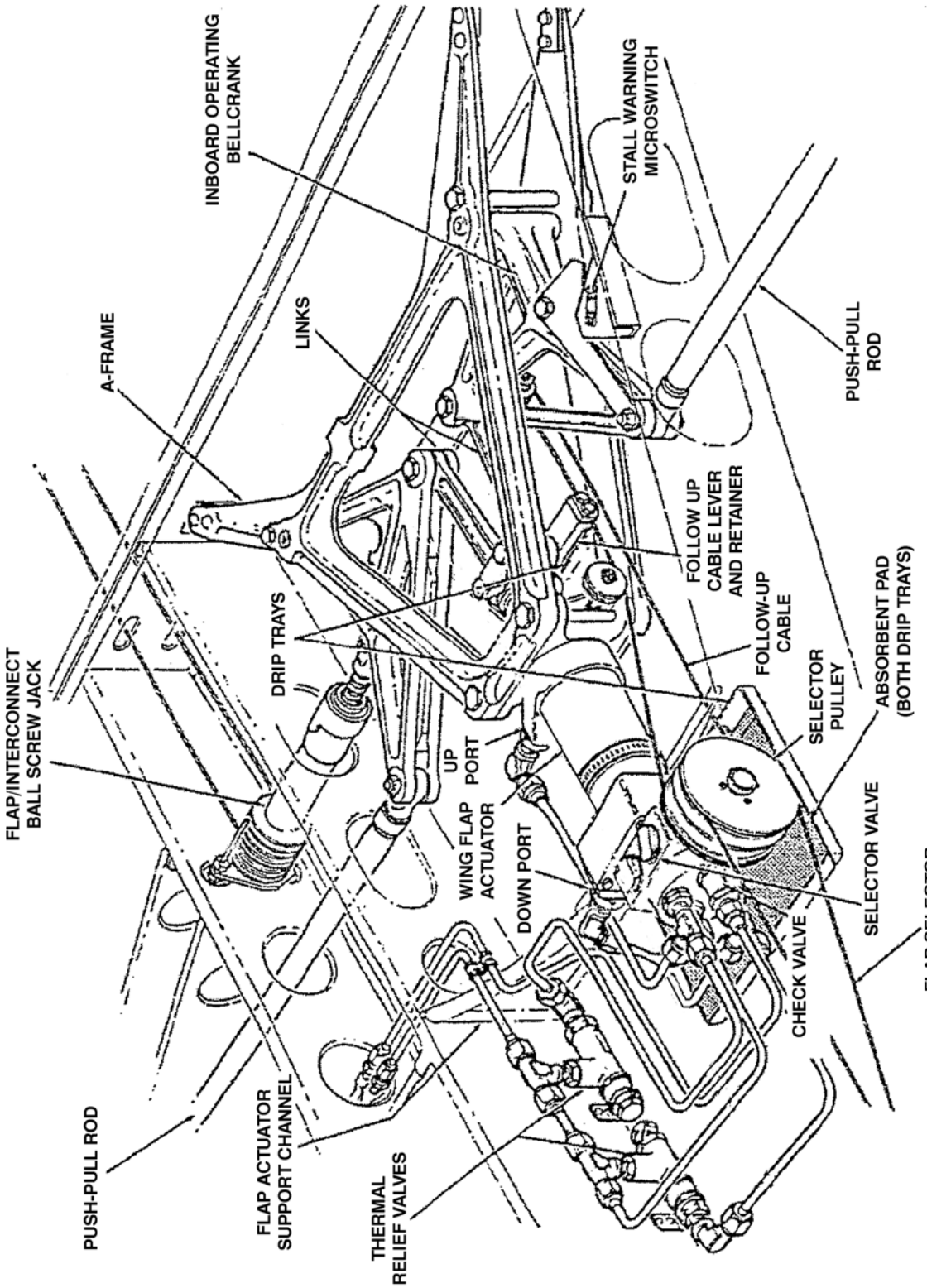
8 FIRST, TIGHTEN BOTH NUTS TO ENSURE A SNUG CONTACT AGAINST THE WASHER/BRACKET. THEN, TIGHTEN BOTH NUTS BY 1 TO 3 CASTELLATED SLOTS TO ALIGN COTTER PIN HOLE ON THE BOLT. IF MORE THAN 50% OF PIN HOLE IS ABOVE THE NUT CASTELLATION, ADD NAS1149D0463K OR NAS1149D0416 WASHER UNDER NUT.

9 MAKE SURE THE ROLLERS/SPACERS ARE TURNING FREELY ON BOLTS (ROD NOT SLIDING) WHILE THE FLAP RODS MOVE IN AND OUT.



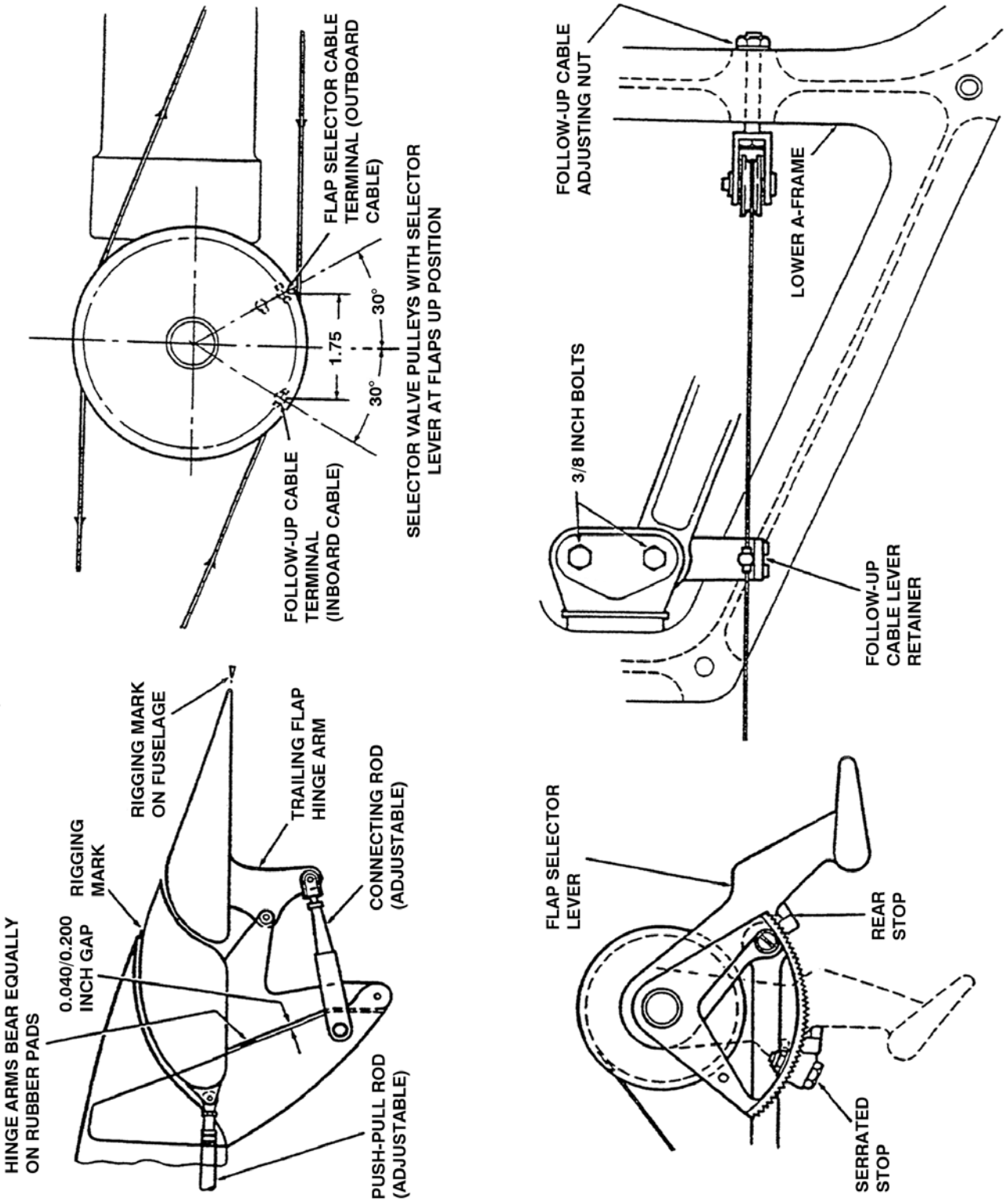
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Wing Flap System
Figure 1 (Sheet 3 of 4)



Wing Flap System
Figure 1 (Sheet 3 of 4)

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Wing Flaps System – Rigging
Figure 2

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2. Adjustment/Test

A. Bench Pressure Test

Refer to Figure 3 for details.

Equipment required:

- Hydraulic test rig capable of supplying and maintaining 2250 psi and flow rate of 1.0 gpm
- Clean hydraulic fluid (MIL–H–5606)

1) Proof pressure:

a) Connect test rig to port P, cap port R, and connect ports C1 and C2 together with shorting line.

b) Apply and maintain 2250 psi hydraulic pressure for two minutes, and rotate pulleys A and B together. Check there is no external leakage, permanent set or deformation.

c) Release hydraulic pressure, remove cap front port R, leave ports C1 and C2 connected by shorting line, rotate pulleys A and B to neutral position, apply 10 psi hydraulic-pressure and maintain for two minutes. Check there is no external leakage.

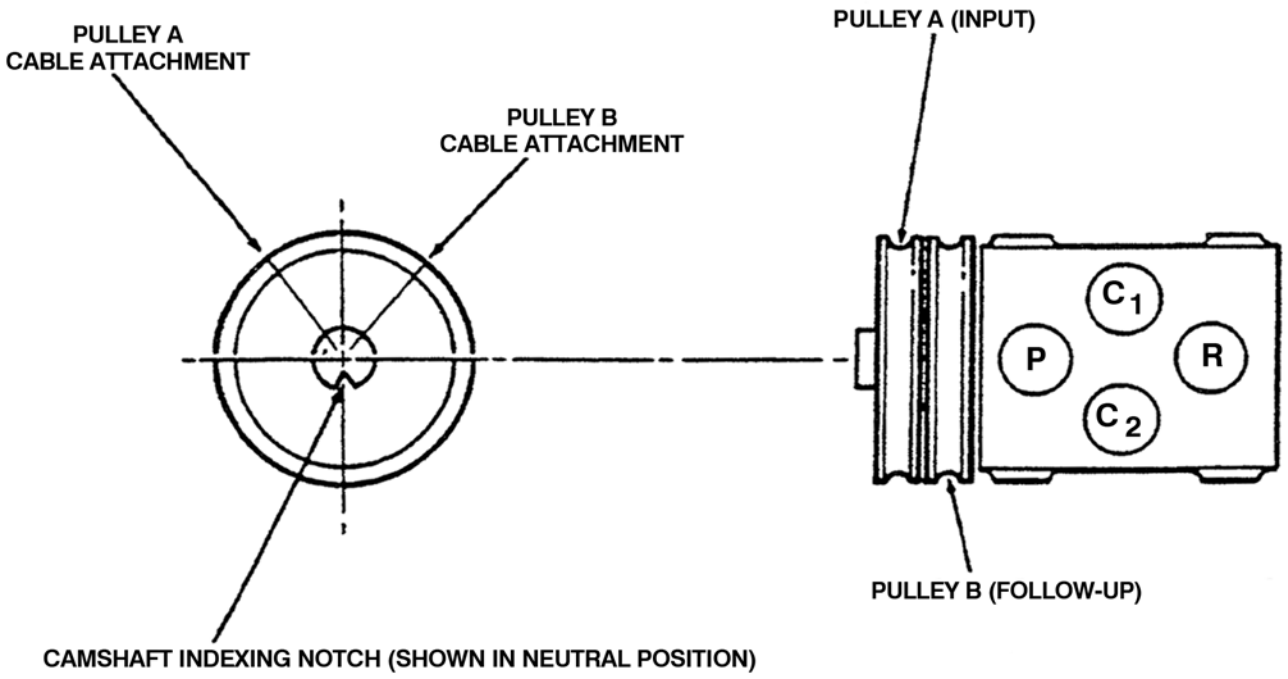
2) Internal leakage:

a) Connect test rig to port P, leave ports R, C1 and C2 open. Ensure pulleys A and B are in neutral position.

b) Apply 1500 psi hydraulic pressure at flow rate of 0.5 to 1.0 gpm, check ports R, C1 and C2 for leaks; leakage must not exceed one drop in three minutes. Release hydraulic pressure then repeat test at 500 psi.

c) Release hydraulic pressure, connect test rig to port R; leave ports P, C1 and C2 open. Ensure pulleys A and B are in neutral position.

d) Apply 1500 psi hydraulic pressure at flow rate of 0.5 to 1.0 gpm, check ports P, C1 and C2 for leaks; leakage must not exceed one drop in three minutes. Release hydraulic pressure then repeat test at 500 psi.



Flap Selector Valve – Test Ports
Figure 3

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